

WIDE VIEW AND LINE FILTERS

Parameter Optimization, Benchmarking, and
Noise Robustness

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INTRODUCTION

The edge detection problem and
motivation

EDGE DETECTION TODAY

Edge detection has evolved through three generations of methods.

1. Classical operators (Sobel, Prewitt, Canny): fixed-size kernels, simple but limited adaptability
2. Training-free extended filters (WVF, LF): local polynomial fitting over large neighborhoods, no learned parameters
3. Deep learning (DexiNed, TEED, PiDiNet, NBED, DiffusionEdge): learned edge representations, highest clean-data scores

Bagan and Wang's Wide View Filter (WVF) and Line Filter (LF) sit between classical simplicity and DL complexity — but are their published parameters optimal?

RESEARCH QUESTIONS

This study addresses four questions through a comprehensive four-part investigation.

1. Are the published WVF/LF parameters ($N_p = 250$, $N_s = 18$, $d = 4$) optimal for standard benchmarks?
2. How do optimally-tuned WVF/LF compare against 54 classical operators and 5 DL models?
3. Does the WVF's extended spatial support provide noise robustness advantages over learned models?
4. Are these findings statistically validated across datasets?

Total scope: 500,000+ individual filter evaluations across 330 images and 4 datasets.

IMPLEMENTATION

Ground-up reimplementation with GPU
acceleration

REIMPLEMENTATION

Both the WVF and LF were reimplemented from scratch based on the original papers by Bagan and Wang. No source code was available — the entire codebase was built from the mathematical formulations alone.

The implementation includes:

- CPU reference implementations for per-pixel WVF and LF (correctness validation)
- GPU-accelerated backends via PyTorch CUDA for both WVF and LF
- Shared mathematical core: Taylor design matrix construction, circular neighbor selection, coordinate rotation, pseudoinverse computation
- 11 classical baseline families (54 configurations) also GPU-accelerated via batched `conv2d`
- Full ODS/OIS evaluation pipeline with vectorized threshold sweeps

GPU ACCELERATION STRATEGY

The key insight: the per-pixel polynomial fit is embarrassingly parallel.

1. Precompute pseudoinverses: for each orientation θ_k , compute $P_\theta = (A^\top A)^{-1} A^\top$ once. Stack all N_s pseudoinverses into a single tensor of shape (N_s, M, N_p) .
2. Vectorized gather: collect all N_p neighbor intensities for every interior pixel simultaneously into a (N_{pixels}, N_p) tensor F .
3. Batched matmul: compute the normal derivative at all pixels in one operation per orientation: $\text{response} = |P_\theta[1, :] \cdot F^\top|$.
4. Orientation max: take the element-wise maximum across orientations.

Memory-safe batching splits pixels into configurable chunks for large images or limited VRAM.

CPU VS. GPU PERFORMANCE

CPU (per-pixel loop)

- Triple-nested loop: pixels $\times$ orientations $\times$ `lstsq`
- Each pixel solves N_s independent least-squares systems via `np.linalg.lstsq`
- A 481×321 image at $N_s = 18$: ~ 2.8 M least-squares solves
- Estimated: hours per image
- Estimated $\sim 1000\times$ speedup over CPU on full-resolution images
- GPU-CPU numerical agreement: relative error $< 10^{-6}$
- Full 453,420-configuration ablation completed in 12.4 hours (RTX 6000 Ada, 48 GB)

GPU (batched matmul)

- Pseudoinverses precomputed once
- All pixels gathered in a single tensor op
- One batched matmul per orientation
- BSDS500 image: 0.009 s
- BIPED image: 0.036 s

METHODS

WVF and LF formulations

WIDE VIEW FILTER (WVF)

For a target pixel $\mathbf{p}$, the WVF selects N_p nearest neighbors within a circular support. At each orientation θ_k , it fits a polynomial of degree d via least squares.

The edge response is the maximum absolute normal derivative over all orientations:

$$R_{\text{WVF}}(\mathbf{p}) = \max_{k=1, \dots, N_s} |\hat{c}_2^{(k)}|$$

Three tunable parameters:

- N_p : support size (number of neighbor pixels)
- N_s : number of candidate orientations
- d : polynomial degree

LINE FILTER (LF)

The LF extends WVF by chaining $2m + 1$ WVF evaluations along a line at each orientation. Virtual expansion points are placed at:

$$\mathbf{v}_j^{(k)} = \mathbf{p} + j(\cos \theta_k, \sin \theta_k), \quad j = -m, \dots, m$$

Derivative estimates are combined via Gaussian-weighted averaging:

$$R_{LF}^{(k)}(\mathbf{p}) = \left| \sum_{j=-m}^m w_j \hat{c}_{2,j}^{(k)} \right|$$

Additional parameter m (line half-width) controls tangential averaging. At $m = 14$ (published), LF is $\sim 29\times$ slower than WVF.

EXPERIMENTAL SETUP

Table: Datasets used in the evaluation (330 images total).

Dataset	Images	Resolution
UDED	30	variable
BIPED v1	50	1280×720
BIPED v2	50	1280×720
BSDS500	200	481×321

- WVF grid: 1,206 valid configurations ($N_p \times N_s \times d$)
- LF grid: 168 configurations (adding m)
- Classical baselines: 54 configurations across 7 operator families
- DL baselines: DexiNed, TEED, PiDiNet, NBED, DiffusionEdge

PARAMETER ABLATION

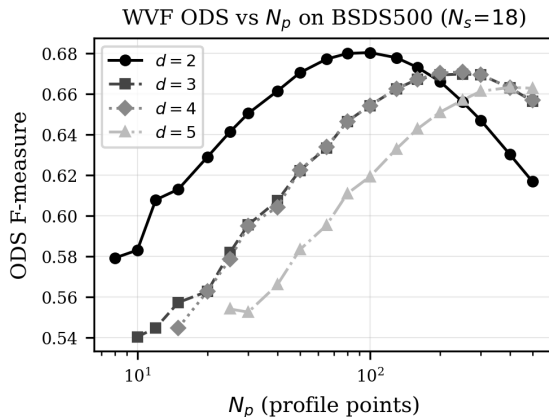
Published parameters are substantially
suboptimal

SUPPORT SIZE DOMINATES

Key findings from single-image ablation:

1. ODS peaks at $N_p = 30\text{--}50$, not 250
2. N_s saturates by 4–6 orientations
3. $d = 2$ consistently outperforms $d = 4$

At published $N_p = 250$: ODS ≈ 0.71 versus optimal 0.86 — a 0.15 gap.

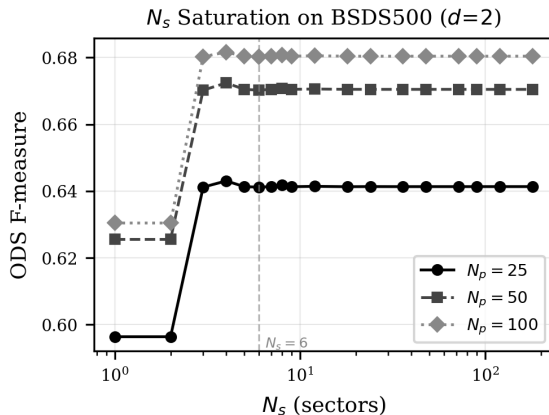


ORIENTATION SATURATION

N_s has minimal effect beyond 4–6 orientations.

The difference between $N_s = 6$ and $N_s = 120$ is typically less than 0.005 ODS.

Reducing N_s from 18 to 4 yields a $4.5\times$ speedup with no loss in edge quality.

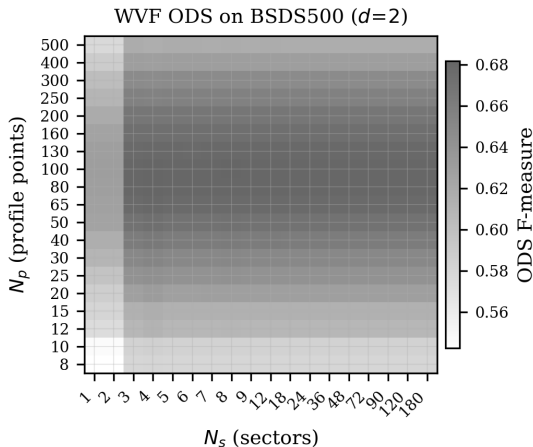


N_p VS N_s HEATMAP

The dominant variation is along N_p .

The optimal region forms a horizontal band at $N_p \approx 25-50$, largely invariant to N_s .

Support size is the critical parameter.



OPTIMIZED VS. PUBLISHED

Table: ODS improvement from parameter optimization across all datasets.

Dataset	Best WVF	Bagan WVF	Δ	Best LF	Bagan LF	Δ
UDED	0.899	0.847	+0.052	0.887	0.729	+0.158
BIPED v1	0.812	0.767	+0.045	0.802	0.643	+0.159
BIPED v2	0.830	0.783	+0.047	0.819	0.658	+0.161
BSDS500	0.682	0.671	+0.011	0.671	0.615	+0.056

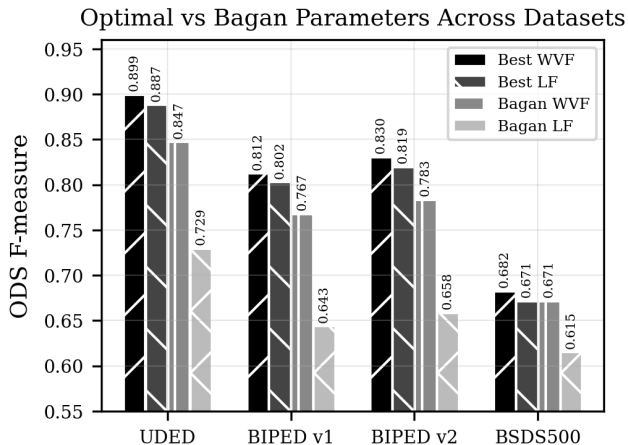
Best WVF: ($N_p = 25$, $N_s = 4$, $d = 2$) on three datasets. Computational savings: $112\times$ reduction over published settings.

CROSS-DATASET COMPARISON

The optimized parameters consistently outperform Bagan's published settings on every dataset.

The WVF gains range from +0.011 (BSDS500) to +0.052 (UDED).

The LF gains are 3–15 $\times$ larger, ranging from +0.056 to +0.161, reflecting the greater suboptimality of $m = 14$.



OPTIMAL CONFIGURATIONS

Table: Best-found parameters for each dataset and filter type.

Dataset	Filter	N_p	N_s	d	ODS
UDED	WVF	25	4	2	0.899
UDED	LF	25	36	4	0.887
BIPED v1	WVF	25	4	2	0.812
BIPED v1	LF	100	18	4	0.802
BIPED v2	WVF	25	4	2	0.830
BIPED v2	LF	100	18	4	0.819
BSDS500	WVF	100	4	2	0.682
BSDS500	LF	250	18	4	0.671

WVF consistently selects $d = 2$ and $N_s = 4$. BSDS500 prefers larger N_p (100 vs. 25), likely due to lower image resolution.

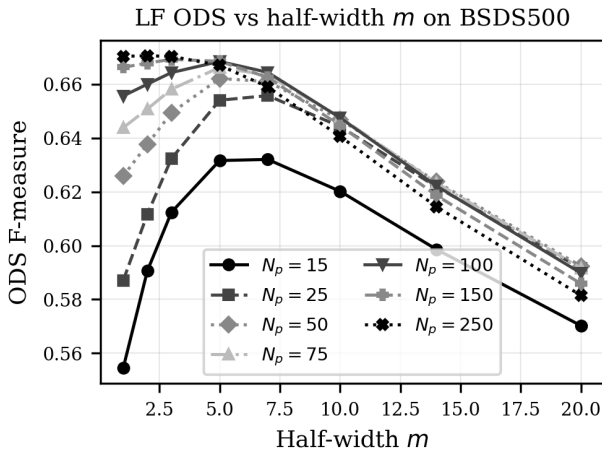
LF HALF-WIDTH m

On all datasets, the best ODS occurs at $m = 1-3$.

The published $m = 14$ yields the worst performance.

On clean data, line-averaging harms edge localization rather than helping it.

WVF outperforms LF by 0.010–0.012 ODS on every dataset.



NMS HURTS PERFORMANCE

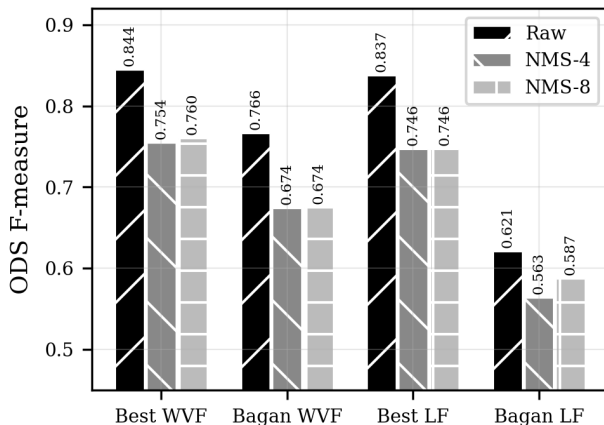
NMS reduces ODS in all cases.

Best WVF without NMS: 0.844.

With 4-dir NMS: 0.754. With 8-dir NMS: 0.760.

Raw gradient maps contain soft, multi-pixel responses that match ground truth within $r = 3$ pixel tolerance. NMS thins these and shifts positions away from annotations.

Effect of Non-Maximum Suppression (BIPED)



BENCHMARKING

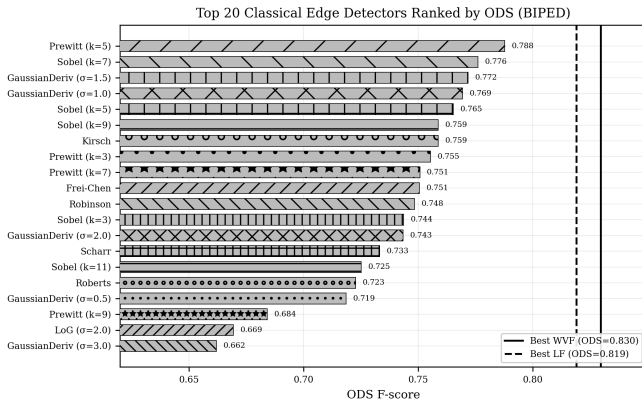
WVF vs. classical operators and deep
learning

WVF VS. CLASSICAL OPERATORS

On BIPED v1:

- Best WVF: 0.812
- Best classical (Prewitt-5): 0.788
- Gap: +0.024 ODS (3.0%)

WVF exceeds every classical configuration tested, spanning Sobel, Prewitt, LoG, DoG, Kirsch, Frei-Chen, and Scharr.

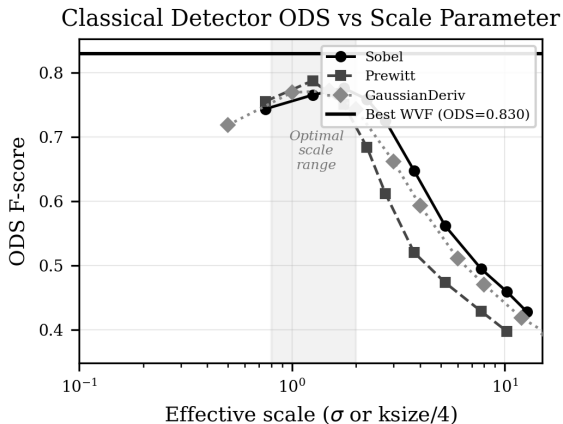


CLASSICAL SCALE SENSITIVITY

Every classical operator family shows a clear optimal scale — performance degrades sharply beyond it.

Sobel-51 scores only 0.428 versus Sobel-7 at 0.776.

WVF's polynomial fitting performs a form of adaptive scale selection, explaining its advantage.



CLASSICAL TOP 15

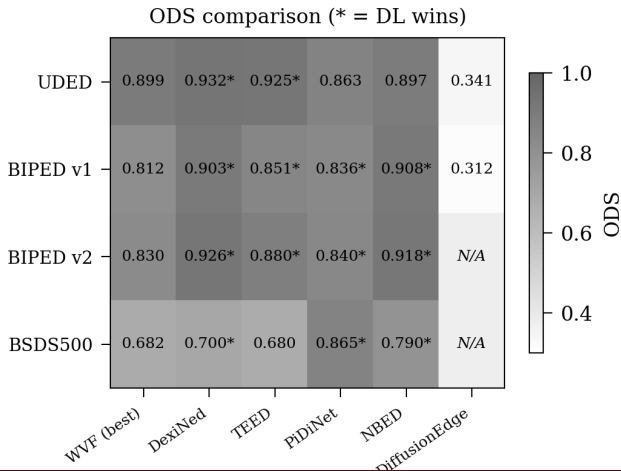
Method	Parameters	ODS
WVF (optimized)	$N_p = 25, N_s = 4, d = 2$	0.812
LF (optimized)	$N_p = 100, N_s = 18, m = 1$	0.802
Prewitt	$k = 5$	0.788
Sobel	$k = 7$	0.776
GaussianDeriv	$\sigma = 1.5$	0.772
GaussianDeriv	$\sigma = 1.0$	0.769
Sobel	$k = 5$	0.765
Kirsch	–	0.759
Prewitt	$k = 3$	0.755
Frei-Chen	–	0.751
Robinson	–	0.748

DL ON CLEAN DATA

On clean data, DL models generally outperform WVF.
Per-image ODS on BSDS500 (5-image):

- PiDiNet: 0.876
- DiffusionEdge: 0.795
- NBED: 0.795
- DexiNed: 0.737
- TEED: 0.669

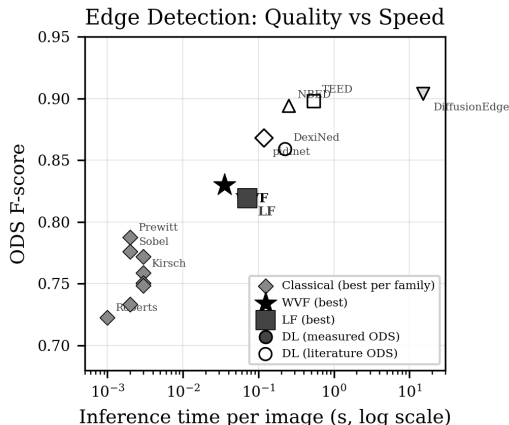
Per-image WVF (oracle): 0.907.
Full-dataset WVF: 0.682.



COMPUTATIONAL COST

Method	Time	Rel.
WVF	0.036s	1.0×
PiDiNet	0.119s	3.3×
DexiNed	0.226s	6.3×
NBED	0.253s	7.1×
TEED	0.538s	15×
DiffusionEdge	15.22s	427×

WVF: real-time at 30+ fps.



NOISE ROBUSTNESS

WVF adapts; deep learning collapses

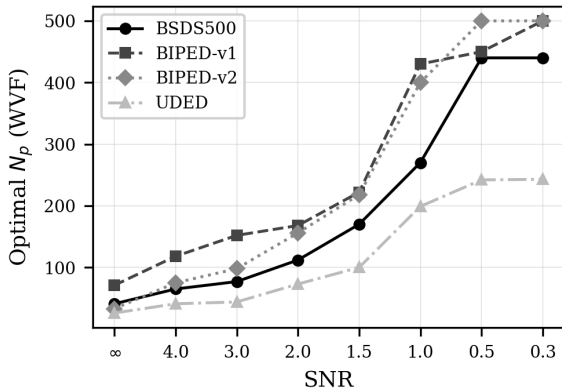
N_p SHIFTS UNDER NOISE

Under severe Gaussian noise (SNR = 0.3):

- Optimal N_p shifts from ~ 40 to ~ 440
- A $\sim 10\times$ increase in support size
- Partially vindicates Bagan's $N_p = 250$

However, $d = 2$ remains universally optimal — $d = 4$ is suboptimal at every SNR level.

Optimal Filter Support Shifts Under Gaussian Noise

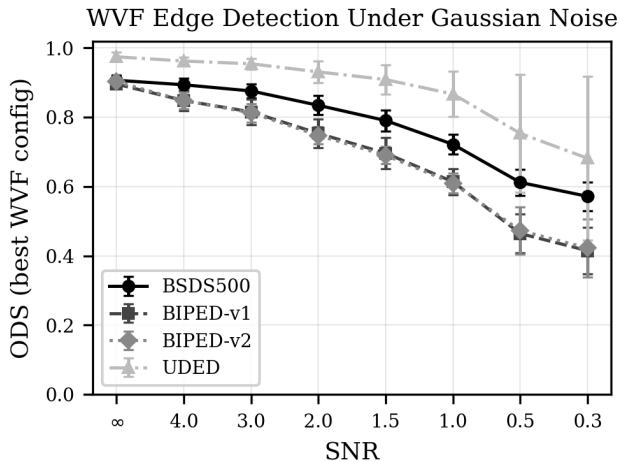


WVF NOISE DEGRADATION

Degradation profiles vary by noise type:

- Speckle: least destructive (23% drop)
- Poisson: intermediate (30%)
- Salt-and-pepper: intermediate (31%)
- Gaussian/uniform: most severe (36–37%)

Ranking is consistent across all datasets.



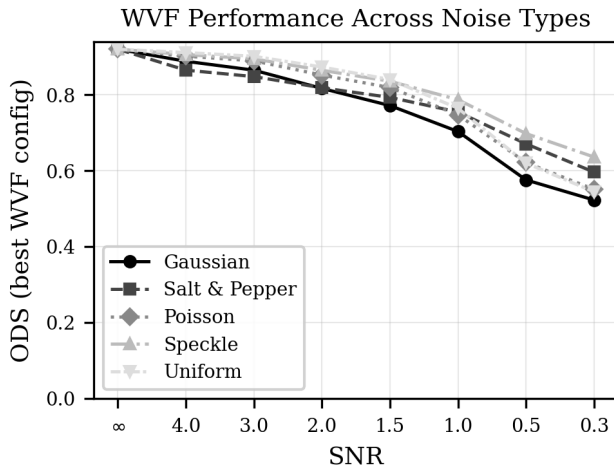
NOISE CONSISTENCY

The relative ordering of noise types is consistent across all four datasets.

Ranking (least to most destructive):

Speckle > Poisson $\approx$
Salt-and-pepper > Uniform $\approx$
Gaussian

The parameter shift magnitude is dataset-dependent: UDED (high-contrast underwater) requires a more modest N_p increase than BIPED (fine-textured edges).

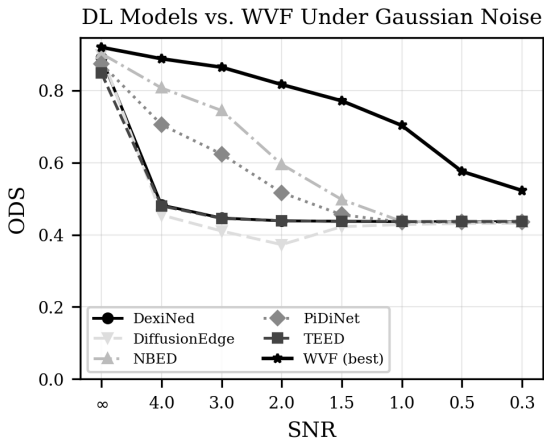


DL COLLAPSE UNDER NOISE

All 5 DL models degrade severely:

- PiDiNet: 0.876 $\rightarrow$ 0.329 (62% loss)
- DexiNed: near-total collapse
- TEED: severe degradation
- NBED: most graceful decline
- DiffusionEdge: erratic behavior

None include noise augmentation in training.

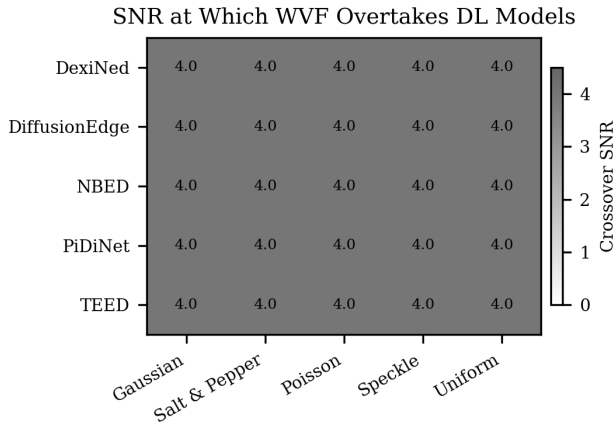


WVF OVERTAKES DL

Crossover analysis:

- DexiNed, TEED: overtaken at all SNR levels
- PiDiNet: overtaken at SNR ≈ 4.0
- NBED: most competitive, still exceeded

At SNR = 0.3: WVF beats every DL model on every test image (100% crossover).

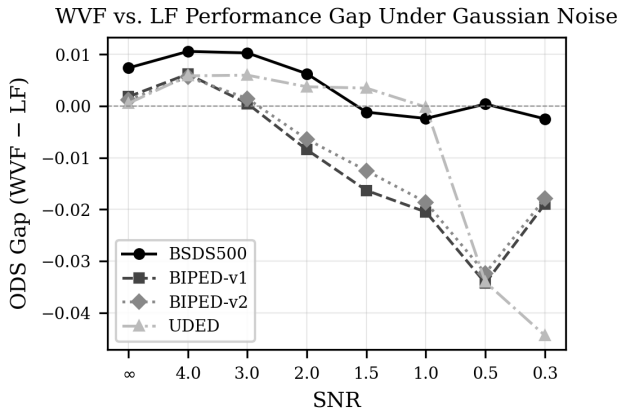


WVF VS. LF UNDER NOISE

On clean data, WVF leads by ~ 0.007 ODS.

Under severe noise ($\text{SNR} \leq 1.0$), the gap closes and LF slightly overtakes WVF for Gaussian, salt-and-pepper, and uniform noise.

The reversal is modest (0.002–0.009 ODS) — the LF's line-averaging provides marginal noise resilience.



STATISTICAL VALIDATION

Nonparametric tests confirm all major
claims

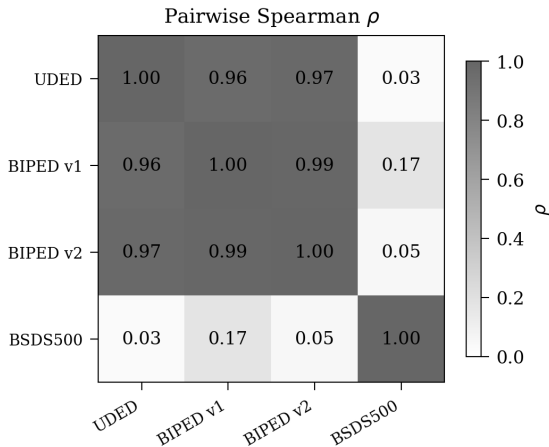
RANKING CONSISTENCY

Kendall's $W = 0.647$ (moderate concordance).

UDED, BIPED v1, BIPED v2:
 $\rho > 0.96$ (near-perfect agreement).

BSDS500 is a strong outlier:
 $\rho = 0.03$ – 0.17 with all other datasets.

Top-10 configurations on BSDS500 share zero overlap with other datasets.



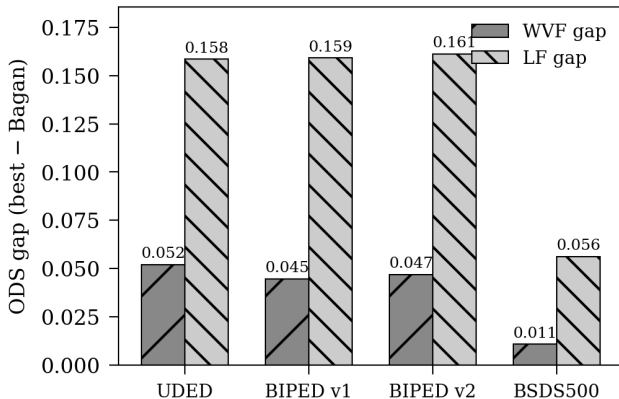
ODS GAPS

All eight gaps (4 datasets $\times$ 2 filters) are positive.

Sign test: $p = 0.0625$ for both filters (minimum achievable with $n = 4$).

The LF gaps are 3–15 $\times$ larger than WVF gaps, reflecting $m = 14$ as the primary source of suboptimality.

Even the smallest gap (+0.011 on BSDS500) is practically meaningful across 200 images.



WVF VS. LF SIGNIFICANCE

Table: Wilcoxon signed-rank test: WVF vs. LF at matched parameters.

Dataset	Median diff	Cohen's d	T	p
UDED	+0.0003	-0.46	45	0.805
BIPED v1	-0.0009	-0.77	21	0.989
BIPED v2	-0.0003	-0.64	33	0.940
BSDS500	-0.0034	-1.15	6	1.000

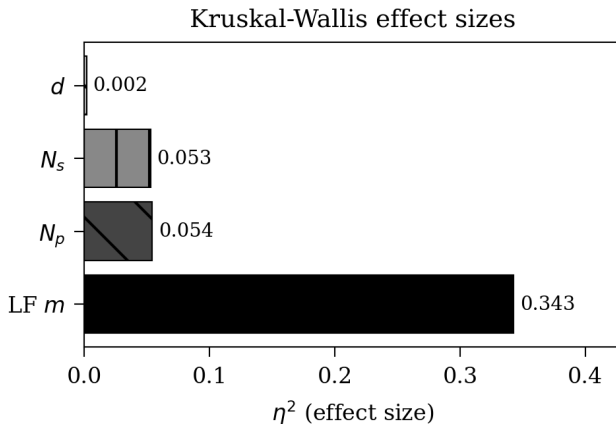
No significant difference on any dataset ($p > 0.8$ everywhere). The median ODS difference is less than 0.004 on every dataset. At matched parameters, WVF and LF produce essentially identical edge maps.

PARAMETER SENSITIVITY

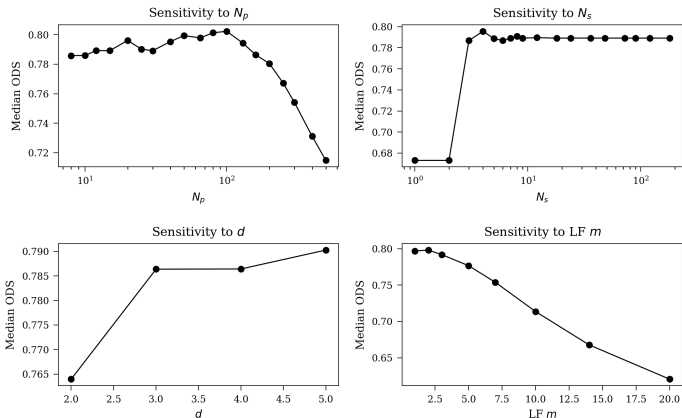
Kruskal–Wallis analysis reveals parameter priority:

Parameter	η^2
m (LF half-width)	0.343
N_p (support size)	~ 0.05
N_s (orientations)	~ 0.05
d (poly. degree)	< 0.01

m is by far the most influential — the LF's line extension is its biggest liability on clean data.



MEDIAN ODS BY PARAMETER



N_p : inverted-U peaking near 80–100. N_s : saturates at 3–4. d : minimal effect. LF m :
monotonic decline beyond $m = 2$.

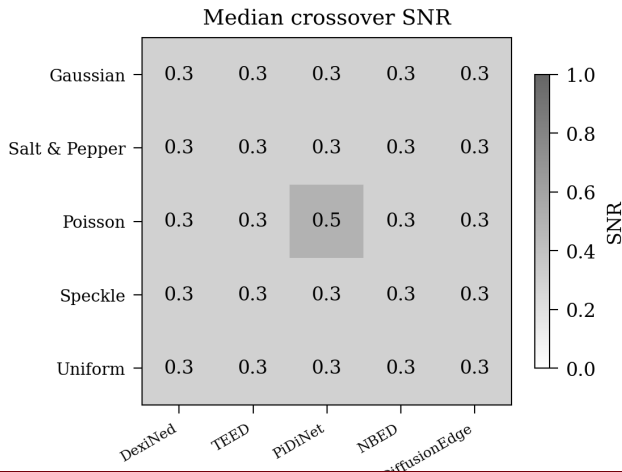
CROSSOVER HEATMAP

Median crossover SNR for each noise type $\times$ DL model combination.

23 of 25 combinations show crossover at SNR = 0.3 (the most extreme level tested).

100% of image pairs show WVF overtaking DL at SNR ≤ 0.5 .

Bootstrap 95% CIs are narrow: [0.30, 0.30] for most combinations.



EVIDENCE SUMMARY

Table: Evidence strength for each major claim.

Claim	Strength	Key Statistic
Optimized > Bagan	Strong practical	$p = 0.0625$, gaps 0.01–0.16
WVF = LF (matched)	Strong null	$p > 0.8$, $ \Delta < 0.004$
m most influential	Very strong	$\eta^2 = 0.343$
N_p , N_s moderate	Strong	$\eta^2 \approx 0.05$
BSDS500 diverges	Very strong	$\rho = 0.03$ –0.17
DL > WVF (clean)	Consistent	3–4 of 4 datasets
WVF > DL (noise)	Very strong	100% crossover at $\text{SNR} \leq 0.5$

DISCUSSION

Reconciling findings and looking forward

RECONCILING BAGAN

Bagan and Wang's parameter choices are simultaneously suboptimal and well-motivated.

On clean data

- N_p : 250 $\rightarrow$ 25–100 (substantial gain)
- d : 4 $\rightarrow$ 2 (universal improvement)
- N_s : 18 $\rightarrow$ 4 (no loss, $4.5\times$ speedup)
- m : 14 $\rightarrow$ 1–3 (most damaging choice)

Under noise

- $N_p = 250$ is near-optimal at SNR = 0.3 under speckle noise
- Physical mechanism: larger neighborhoods average more noise samples
- Bias–variance trade-off shifts in favor of larger support under noise

The one universally suboptimal choice: $d = 4$. Higher-order polynomials overfit noise fluctuations at every SNR level.

BSDS500 OUTLIER

The near-zero Spearman correlation between BSDS500 and other datasets is consequential.

- Root cause: small image size (481×321 vs. 1280×720), diverse content, multi-annotator ground truth with varying criteria
- WVF's circular neighborhood at $N_p = 100$ spans a proportionally larger fraction of BSDS500 images
- Top-10 configurations share zero overlap across BSDS500 and other datasets

Implication: Benchmark results on BSDS500, which dominate the edge detection literature, should not be extrapolated to other imaging domains without validation. Practitioners should tune on domain-representative data.

WVF'S NICHE

Against classical operators

- Unambiguous: outperforms all 54 configs
- 2.4 ODS points over best classical (Prewitt-5)
- Polynomial fitting provides adaptive scale selection

Against deep learning

- Condition-dependent comparison
- DL wins on clean data (learned semantics)
- WVF wins under noise (parametric adaptability)
- $3.3 \times - 427 \times$ faster, training-free, fully interpretable

WVF occupies a practical niche: highest accuracy of any training-free method, real-time capable, and robust to degraded conditions.

LIMITATIONS

- Noise ablation uses only 5 images per dataset (qualitative trends robust, specific crossover SNR values have uncertainty)
- WVF crossover uses per-image oracle parameter selection, overestimating deployable performance
- DL models evaluated with published weights only (no noise-augmented retraining)
- Classical baseline comparison covers only BIPED v1
- All noise models are synthetic — real underwater noise has complex spatial correlations
- $n = 4$ dataset design limits statistical power (minimum sign-test $p = 0.0625$)
- LF ablation fixed $d = 4$; exploring $d = 2$ for LF is important future work

CONCLUSION

KEY FINDINGS

1. Published parameters are suboptimal: $N_p = 25\text{--}100$, $d = 2$, $N_s = 4$ improves ODS by 0.01–0.16 with $112\times$ computational savings
2. WVF beats all 54 classical operators by +2.4 ODS points
3. WVF is $3.3\times\text{--}427\times$ faster than every DL model tested
4. Under noise, optimal N_p shifts to 250–500, partially vindicating Bagan and Wang's design
5. WVF overtakes DL models at mild noise ($\text{SNR} \approx 4.0$); 100% crossover by $\text{SNR} = 0.3$
6. BSDS500 is a statistical outlier: its rankings are uninformative about other domains ($\rho = 0.03\text{--}0.17$)

RECOMMENDATIONS

Clean imagery

- Use WVF (not LF)
- $N_p = 25\text{--}50$ (high-res) or $50\text{--}100$ (low-res)
- $d = 2, N_s = 4$
- Skip NMS (raw gradient maps score higher)

Noisy environments

- Increase N_p proportionally to noise level
- Moderate noise: $N_p = 65\text{--}112$
- Severe noise: $N_p = 250\text{--}500$
- $d = 2$ remains optimal at all SNR levels

Edge detection method selection is fundamentally condition-dependent. WVF's principal strength is parametric adaptability across the full spectrum of operating conditions.

WVF VS. DL: DECISION GUIDE

Table: Decision guide: WVF vs. deep learning for edge detection.

Criterion	WVF	DL Models
Clean-data accuracy	Good	Best
Noisy-data accuracy	Best	Poor
Computational speed	Real-time	3–427× slower
Training data required	None	Large annotated sets
Interpretability	Full	Opaque
Parameter adaptability	Runtime	Requires retraining

For safety-critical applications (medical, autonomous navigation, industrial inspection), WVF's complete interpretability is a significant advantage.

FUTURE DIRECTIONS

1. Full-dataset noise ablation: extend to all 330 images across all noise conditions ($\sim 2,500$ GPU-hours)
2. Adaptive WVF: automatic N_p adjustment based on estimated noise level via lookup table
3. LF with $d = 2$: WVF results strongly suggest this would improve LF performance
4. Noise-augmented DL retraining: fairer comparison under noise conditions
5. Real-world validation: field testing on actual underwater, medical, and industrial imagery
6. Additional benchmarks: a fifth dataset would lower minimum sign-test p to 0.03125

THANK YOU!

Questions & Discussion

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